

1	Suggested Solution
(i)	Let the 4 distinct positive roots be p, q, r and s. Then $f(x) = (x-p)(x-q)(x-r)(x-s)$ $= x^4 - x^3(p+q+r+s) + x^2(pq+pr+ps+qr+qs+rs) - x(pqr+pqs+qrs+prs) + pqrs$ Comparing the expression for $f(x)$, we have $pqr+pqs+qrs+prs=4c^3, \quad pqrs=d^4.$ Using AM-GM inequality, we get $c^3 = \frac{pqr+pqs+qrs+prs}{4}$ $> \sqrt[4]{(pqr)(pqs)(qrs)(prs)} = \sqrt[4]{(pqrs)^3} = \sqrt[4]{d^{12}} = d^3$ Hence $c > d$.
(ii)	We know that $y=f(x)$ has 3 turning points, i.e. $f'(x) = 4x^3 - 12ax^2 + 12b^2x - 4c^3 = 0$ has 3 distinct real solutions, namely t_1, t_2 and t_3. $0 = x^3 - 3ax^2 + 3b^2x - c^3$ $= (x-t_1)(x-t_2)(x-t_3)$ $= x^3 - x^2(t_1+t_2+t_3) + x(t_1t_2+t_1t_3+t_2t_3) - t_1t_2t_3$ Similar to the previous part, using AM-GM inequality yields $b^2 = \frac{t_1t_2+t_1t_3+t_2t_3}{3} > \sqrt[3]{(t_1t_2)(t_1t_3)(t_2t_3)} = \sqrt[3]{(t_1t_2t_3)^2} = \sqrt[3]{c^6} = c^2.$ Hence $b > c$.

2	Suggested Solution
(i)	$F_2 = F_1 + F_0 = 1$ $F_3 = F_2 + F_1 = 2$ $F_4 = F_3 + F_2 = 3$ $F_5 = F_4 + F_3 = 5$
	$n = 1: F_2F_0 - F_1^2 = -1$ $n = 2: F_3F_1 - F_2^2 = 1$ $n = 3: F_4F_2 - F_3^2 = -1$ Conjecture: $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$ Let P_n denote "$F_{n+1}F_{n-1} - F_n^2 = (-1)^n$", for $n \in \mathbb{Z}^+$. When $n = 1$, LHS = $F_2F_0 - F_1^2 = -1$; RHS = $(-1)^1 = -1$ Hence P_1 is true. Suppose P_k are true for some $k \in \mathbb{Z}^+$, i.e. $F_{k+1}F_{k-1} - F_k^2 = (-1)^k.$ We want to prove that $F_{k+2}F_k - F_{k+1}^2 = (-1)^{k+1}$. LHS of P_{k+1} $ \begin{aligned} &= F_{k+2}F_k - F_{k+1}^2 \\ &= (F_{k+1} + F_k)F_k - F_{k+1}^2 \\ &= F_{k+1}F_k + F_k^2 - F_{k+1}^2 \\ &= F_{k+1}F_k + F_{k+1}F_{k-1} - (-1)^k - F_{k+1}^2 \\ &= F_{k+1}(F_k + F_{k-1}) - (-1)^k - F_{k+1}^2 \\ &= F_{k+1}^2 + (-1)^{k+1} - F_{k+1}^2 = (-1)^{k+1} \end{aligned} $ Hence P_k is true implies that P_{k+1} is true. Since P_1 is true as well, by induction, P_n is true for $n \in \mathbb{Z}^+$.

(iii)

Let Q_k denote " $F_{n+k} = F_k F_{n+1} + F_{k-1} F_n$ ", for $n, k \in \mathbb{Z}^+$.

When $k = 1$, LHS = F_{n+1} ; RHS = $F_1 F_{n+1} + F_0 F_n = F_{n+1}$.

When $k = 2$,

LHS = F_{n+2} ; RHS = $F_2 F_{n+1} + F_1 F_n = F_{n+1} + F_n = F_{n+2}$

Hence Q_1 and Q_2 are true.

Suppose Q_{j-1} and Q_j are true for some $j \in \mathbb{Z}^+$, i.e.

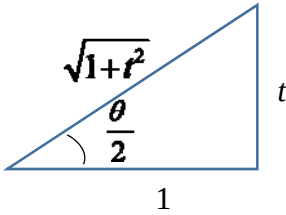
$F_{n+j-1} = F_{j-1} F_{n+1} + F_{j-2} F_n$ and $F_{n+j} = F_j F_{n+1} + F_{j-1} F_n$.

We want to prove that $F_{n+j+1} = F_{j+1} F_{n+1} + F_j F_n$.

$$\begin{aligned} \text{LHS} &= F_{n+j+1} \\ &= F_{n+j-1} + F_{n+j} \\ &= F_{j-1} F_{n+1} + F_{j-2} F_n + F_j F_{n+1} + F_{j-1} F_n \\ &= F_{n+1} (F_{j-1} + F_j) + F_n (F_{j-2} + F_{j-1}) \\ &= F_{j+1} F_{n+1} + F_j F_n = \text{RHS} \end{aligned}$$

Hence Q_{j-1} and Q_j are true imply that Q_{j+1} is true.

Since Q_1 and Q_2 are true as well, by induction, Q_k is true for $n, k \in \mathbb{Z}^+$.

3	Suggested Solution
(i)	Since $t = \tan\left(\frac{\theta}{2}\right)$, we have  $\cos \theta = 2 \cos^2\left(\frac{\theta}{2}\right) - 1$ $= 2\left(\frac{1}{1+t^2}\right) - 1 = \frac{2-1-t^2}{1+t^2} = \frac{1-t^2}{1+t^2}$ $\sin \theta = 2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) = 2\left(\frac{t}{\sqrt{1+t^2}}\right)\left(\frac{1}{\sqrt{1+t^2}}\right) = \frac{2t}{1+t^2}$ Then $\frac{1+\cos \theta}{\sin \theta} = \frac{\left(1+\frac{1-t^2}{1+t^2}\right)}{\left(\frac{2t}{1+t^2}\right)} = \frac{2}{2t} = \frac{1}{t} = \tan\left(\frac{\pi}{2} - \frac{\theta}{2}\right)$ Or $\frac{1+\cos \theta}{\sin \theta} = \frac{1+2\cos^2\frac{\theta}{2}-1}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}} = \cot\frac{\theta}{2} = \frac{1}{t} = \tan\left(\frac{\pi}{2} - \frac{\theta}{2}\right)$
(ii)	Differentiating $t = \tan\left(\frac{\theta}{2}\right)$ w.r.t. θ, we get $\frac{dt}{d\theta} = \frac{1}{2} \sec^2\left(\frac{\theta}{2}\right) = \frac{1}{2\cos^2\left(\frac{\theta}{2}\right)}.$

	$ \begin{aligned} I &= \int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos \alpha \sin \theta} d\theta \\ &= \int_0^1 \frac{1}{1 + \cos \alpha \sin \theta} 2 \cos^2 \left(\frac{\theta}{2} \right) dt \\ &= \int_0^1 \frac{2}{1 + \cos \alpha \left(\frac{2t}{1+t^2} \right)} \left(\frac{1}{1+t^2} \right) dt \\ &= \int_0^1 \frac{2(1+t^2)}{1+t^2 + 2t \cos \alpha} \left(\frac{1}{1+t^2} \right) dt \\ &= \int_0^1 \frac{2}{1+t^2 + 2t \cos \alpha} dt \\ &= \int_0^1 \frac{2}{(t + \cos \alpha)^2 + 1 - \cos^2 \alpha} dt \\ &= \int_0^1 \frac{2}{(t + \cos \alpha)^2 + \sin^2 \alpha} dt \end{aligned} $
(iii)	Differentiating $t + \cos \alpha = \sin \alpha \tan u$ w.r.t. u, we get $\frac{dt}{du} = \sin \alpha \sec^2 u = \frac{\sin \alpha}{\cos^2 u}.$ $ \begin{aligned} &\int_0^1 \frac{2}{(t + \cos \alpha)^2 + \sin^2 \alpha} dt \\ &= \int_{\pi/2-\alpha}^{\pi/2-\alpha/2} \frac{2}{\sin^2 \alpha \tan^2 u + \sin^2 \alpha} \left(\frac{\sin \alpha}{\cos^2 u} \right) du \\ &= \int_{\pi/2-\alpha}^{\pi/2-\alpha/2} \frac{2}{\sin \alpha (\sec^2 u)} \left(\frac{1}{\cos^2 u} \right) du \\ &= \int_{\pi/2-\alpha}^{\pi/2-\alpha/2} \frac{2}{\sin \alpha} du \\ &= \frac{2}{\sin \alpha} \left[\left(\frac{\pi}{2} - \frac{\alpha}{2} \right) - \left(\frac{\pi}{2} - \alpha \right) \right] \\ &= \frac{\alpha}{\sin \alpha} \end{aligned} $
(iv)	Let $\theta = \frac{\pi}{2} - \phi$. Then we have

	$\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \alpha \cos \theta} d\theta = \int_{\frac{\pi}{2}}^0 \frac{-1}{1 + \sin \alpha \cos \left(\frac{\pi}{2} - \phi \right)} d\phi$ $= \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \alpha \sin \phi} d\phi$ $= \int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos \left(\frac{\pi}{2} - \alpha \right) \sin \phi} d\phi$ $= \frac{\frac{\pi}{2} - \alpha}{\sin \left(\frac{\pi}{2} - \alpha \right)} = \frac{\pi - 2\alpha}{2 \cos \alpha}$
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4	Suggested Solution
	$\left(1 + \frac{x}{n} \right)^n = 1 + n \left(\frac{x}{n} \right) + \frac{n(n-1)}{2!} \left(\frac{x}{n} \right)^2 + \frac{n(n-1)(n-2)}{3!} \left(\frac{x}{n} \right)^3 + \dots$ $+ \frac{n(n-1)(n-2)\dots(1)}{n!} \left(\frac{x}{n} \right)^n$ $= 1 + x + \frac{n-1}{n} \left(\frac{x^2}{2!} \right) + \frac{(n-1)(n-2)}{n^2} \left(\frac{x^3}{3!} \right) + \dots + \frac{(n-1)!}{n^{n-1}} \left(\frac{x^n}{n!} \right)$ $a_2 = \frac{n-1}{n}$ $a_3 = \frac{(n-1)(n-2)}{n^2}$ $a_r = \frac{(n-1)(n-2)\dots(n-r+1)}{n^{r-1}}$ $= \left[\frac{(n-1)}{n} \right] \left[\frac{(n-2)}{n} \right] \dots \left[\frac{(n-r+1)}{n} \right]$ $= \left(1 - \frac{1}{n} \right) \left(1 - \frac{2}{n} \right) \dots \left(1 - \frac{r-1}{n} \right)$
	$a_n = \frac{(n-1)(n-2)\dots(1)}{n^{n-1}}$ $= \frac{(n-1)!}{n^{n-1}}$
	For $2 \leq r \leq n$, $\left(\frac{1}{n} \right), \left(\frac{2}{n} \right), \dots, \left(\frac{r-1}{n} \right)$ have values between 0 and 1.

	Hence $0 < \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{r-1}{n}\right) < 1$ $0 < a_r < 1$ (shown)
	For $2 < r \leq n$, $\frac{1}{r!} = \frac{1}{\overbrace{r(r-1)(r-2)\dots 3.2}^{(r-1) \text{ no. of terms}}} < \frac{1}{\overbrace{(2)(2)\dots(2)}^{(r-1) \text{ no. of terms}}} = \frac{1}{2^{r-1}}.$ For $r = 2$, $\frac{1}{2!} = \frac{1}{2^{2-1}} = \frac{1}{2}$ Hence $\frac{1}{r!} \leq \frac{1}{2^{r-1}}.$
	$\begin{aligned} \left(1 + \frac{x}{n}\right)^n &= 1 + x + a_2 \left(\frac{x^2}{2!}\right) + a_3 \left(\frac{x^3}{3!}\right) + \dots + a_n \left(\frac{x^n}{n!}\right) \\ &= 1 + x + \left(\frac{a_2}{2!}\right)x^2 + \left(\frac{a_3}{3!}\right)x^3 + \dots + \left(\frac{a_n}{n!}\right)x^n \\ &< 1 + x + \frac{x^2}{2} + \frac{x^3}{2^2} + \dots + \frac{x^n}{2^{n-1}} \quad \text{since } 0 < a_r < 1 \text{ and } \frac{1}{r!} \leq \frac{1}{2^{r-1}} \\ &\quad \text{for } 2 \leq r \leq n. \end{aligned}$
	$\begin{aligned} \left(1 + \frac{x}{n}\right)^n &< 1 + x + \frac{x^2}{2} + \frac{x^3}{2^2} + \dots + \frac{x^n}{2^{n-1}} \\ &< 1 + \frac{x}{\left(1 - \frac{x}{2}\right)} \quad \text{since } \frac{x^n}{2^{n-1}} > 0 \text{ when } 0 < x < 2 \text{ for all } n \in \mathbf{Z}^+ \\ &= \frac{\left(1 + \frac{x}{2}\right)}{\left(1 - \frac{x}{2}\right)} = \frac{2+x}{2-x} \end{aligned}$ OR

	$\left(1 + \frac{x}{n}\right)^n < 1 + x + \frac{x^2}{2} + \frac{x^3}{2^2} + \dots + \frac{x^n}{2^{n-1}}$ $< 1 + x + \frac{x^2}{2} + \frac{x^3}{2^2} + \dots \quad \text{since } \frac{x^n}{2^{n-1}} > 0$ $= 1 + \frac{x}{\left(1 - \frac{x}{2}\right)} \quad \text{since } r = \left \frac{x}{2}\right < 1 \text{ for } 0 < x < 2$ $= \frac{\left(1 + \frac{x}{2}\right)}{\left(1 - \frac{x}{2}\right)} = \frac{2+x}{2-x}$
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5	Suggested Solution
(a) (i)	Let $P(r)$ be the statement $S(r, 3) = \frac{1}{2}(3^{r-1} - 2^r + 1)$, for $r \geq 3$. For $r = 3$, (LHS) $S(3, 3) = 1$ (RHS) $\frac{1}{2}(3^{3-1} - 2^3 + 1) = \frac{1}{2}(2) = 1$ $\therefore P(3)$ is true. Assume $P(k-1)$ is true for some integer $k-1$, that is, $S(k-1, 3) = \frac{1}{2}(3^{k-2} - 2^{k-1} + 1)$. Prove $P(k)$ is true. $\begin{aligned} S(k, 3) &= S(k-1, 2) + 3S(k-1, 3) \\ &= 2^{k-2} - 1 + \frac{3}{2}(3^{k-2} - 2^{k-1} + 1) \end{aligned}$ For $r = k$, $\begin{aligned} &= \frac{1}{2}(2^{k-1} - 2 + 3 \cdot 3^{k-2} - 3 \cdot 2^{k-1} + 3) \\ &= \frac{1}{2}(3^{k-1} - 2^k + 1) \end{aligned}$ $\therefore P(k)$ is true. Since $P(3)$ is true and $P(k-1)$ is true implies $P(k)$ is true, by Mathematical Induction, $P(r)$ is true for $r \geq 3$.
(a)(i)	Number of wanted ways

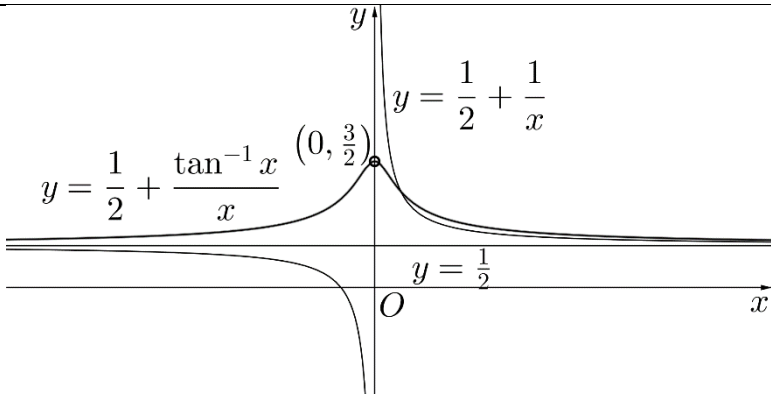
(i)	$= 3 \times S(10, 3)$ $= 55980$
(b)	$a_4 = 1 + 1 + 1 + 1 + 1 = 5$ Suppose there is additional type of tile 1×3, $a_1 = 1, a_2 = 3, a_3 = 6$, Case 1: To pave $1 \times n$ rectangular wall with 1×1 tile, a_{n-1} ways to pave the remaining $1 \times (n-1)$ wall tiles. Case 2: To pave $1 \times n$ rectangular wall with 1×2 tile a_{n-2} ways to pave the remaining $1 \times (n-2)$ wall tiles and there are 2 ways to select the 1×2 tile. Case 3: To pave $1 \times n$ rectangular wall with 1×3 tile, a_{n-3} ways to pave the remaining $1 \times (n-3)$ wall tiles By Addition Principle, $a_n = a_{n-1} + 2a_{n-2} + a_{n-3}, n \geq 4, a_1 = 1, a_2 = 3, a_3 = 6$

6	Suggested Solution
(i) (a)	Number of designs $= 10^4$ $= 10000$
(i) (b)	Number of designs with reflection symmetry in a horizontal line $= 10(10)$ $= 100$
(i) (c)	Let H denote the set of bag design with horizontal symmetry. Let V denote the set of bag design with vertical symmetry. Let R denote the set of bag design with rotational symmetry. $ \overline{H \cap V \cap R} = \overline{H \cup V \cup R} $ $= S - H \cup V \cup R $ $= S - [H + V + R - H \cap V - V \cap R - H \cap R + H \cap V \cap R]$ $= 10^4 - [10^2 + 10^2 + 10^2 - 10 - 10 - 10 + 10]$ $= 9720$
(ii)	Case 1: When 4 colours are used

	The number of patterns is $\frac{1}{4} \binom{c}{4} (4!)$ Case 2: When 3 colours are used (with 2 quadrants having the same colour) The number of patterns is $\frac{1}{4} \binom{c}{3} \binom{3}{1} \frac{4!}{2!}$ Case 3a: When 2 colours are used (with each colour appears are used for exactly 2 quadrants) The number of patterns is $\frac{1}{2} \binom{c}{2} \frac{4!}{2!2!}$ Case 3b: When 2 colours are used (with 3 quadrants having the same colour) The number of patterns is $\frac{1}{4} \binom{c}{2} \binom{2}{1} \frac{4!}{3!}$ Case 4: When only 1 colour is used The number of patterns is $\binom{c}{1}$ So the total number of distinct patterns is $\begin{aligned} & \frac{1}{4} \binom{c}{4} (4!) + \frac{1}{4} \binom{c}{3} \binom{3}{1} \frac{4!}{2!} + \frac{1}{2} \binom{c}{2} \frac{4!}{2!2!} + \frac{1}{4} \binom{c}{2} \binom{2}{1} \frac{4!}{3!} + \binom{c}{1} \\ &= \frac{1}{4} c(c-1)(c-2)(c-3) + \frac{3}{2} c(c-1)(c-2) + \frac{3}{2} c(c-1) + c(c-1) + c \\ &= \frac{1}{4} c[(c^3 - 6c^2 + 11c - 6) + (6c^2 - 18c + 12) + (6c - 6) + (4c - 4) + 4] \\ &= \frac{1}{4} c(c^3 + 3c) \\ &= \frac{1}{4} c^2(c^2 + 3) \end{aligned}$
	Let S denote the number of ways Terry can buy any design for his 8 friends. Thus $S = 5^8$. Let A_i denote the event that Terry did not buy the bag of design i, where $1 \leq i \leq 5$ for his friends. $ A_1 = A_2 = A_3 = A_4 = A_5 = 4^8$ $\sum_{i=1}^5 A_i = \binom{5}{1} 4^8$ $\sum_{1 \leq i < j \leq 5} A_i \cap A_j = \binom{5}{2} 3^8$

	$\sum_{1 \leq i < j < k \leq 5} A_i \cap A_j \cap A_k = \binom{5}{3} 2^8$ $\sum_{1 \leq i < j < k < l \leq 5} A_i \cap A_j \cap A_k \cap A_l = \binom{5}{4} 1^8$ By principle of inclusion and exclusion, number of ways is $5^8 - \binom{5}{1} 4^8 + \binom{5}{2} 3^8 - \binom{5}{3} 2^8 + \binom{5}{4} 1^8 = 126000$
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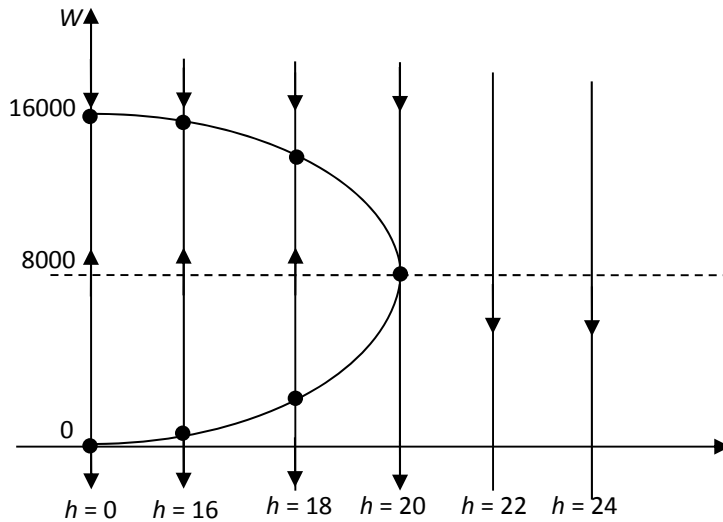
7	Suggested Solution
(a)	Beginning with $x_0 = 0, y_0 = 0.1$, the first two approximate values are $y_1^* = 1 + (0.5) \cdot (1)(0+1) = 1.5$ $y_1 = 1 + (0.5) \cdot \left[\frac{(1)(0+1) + (1.5)(0.5+1.5)}{2} \right] = 2$ $y_2^* = 2 + (0.5) \cdot (2)(0.5+2) = 4.5$ $y_2 = 2 + (0.5) \cdot \left[\frac{(2)(0.5+2) + (4.5)(1+4.5)}{2} \right]$ $= 9.4375$
(b) (i)	$z = x \frac{dy}{dx} + y \Rightarrow \frac{dz}{dx} = x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx}$ $x(1+x^2) \frac{d^2y}{dx^2} + 2(1+2x^2) \frac{dy}{dx} + x(2y-1) = 0$ $x(1+x^2) \frac{d^2y}{dx^2} + 2(1+x^2) \frac{dy}{dx} + 2x^2 \frac{dy}{dx} + 2xy - x = 0$ $(1+x^2) \left(x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} \right) + 2x \left(x \frac{dy}{dx} + y \right) = x$ $(1+x^2) \frac{dz}{dx} + 2xz = x$ $\frac{dz}{dx} + \frac{2x}{1+x^2} z = \frac{x}{1+x^2}$ Integrating factor = $e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = 1+x^2$

	$\frac{dz}{dx} + \frac{2x}{1+x^2}z = \frac{x}{1+x^2}$ $\Rightarrow (1+x^2)\frac{dz}{dx} + 2xz = x$ $\Rightarrow \frac{d}{dx}[(1+x^2)z] = x$ $\Rightarrow (1+x^2)z = \int x \, dx = \frac{x^2}{2} + A$ $\Rightarrow z = \frac{x^2 + 2A}{2(1+x^2)}$ $\Rightarrow x\frac{dy}{dx} + y = \frac{x^2 + 2A}{2(1+x^2)}$ $\Rightarrow \frac{d}{dx}(xy) = \frac{x^2 + 2A}{2(1+x^2)} = \frac{1}{2} - \frac{1-2A}{2(1+x^2)}$ $\Rightarrow xy = \int \left[\frac{1}{2} - \frac{1-2A}{2(1+x^2)} \right] dx = \frac{1}{2}x + B \tan^{-1} x + C, \quad B = \frac{2A-1}{2}$ $\Rightarrow y = \frac{1}{2} + \frac{B \tan^{-1} x + C}{x}$
(b) (ii)	 The graph shows the function $y = \frac{1}{2} + \frac{\tan^{-1} x}{x}$. The curve has a vertical asymptote at $x=0$ and a horizontal asymptote at $y = \frac{1}{2}$. The curve passes through the point $(0, \frac{3}{2})$. The origin is labeled O.

8	Suggested Solution
(i)	Let s represents the spring constant. $5(9.8) = 0.4s \Rightarrow s = 122.5$ Motion of the body: $5 \frac{d^2x}{dt^2} = -k \frac{dx}{dt} - 122.5x$ $\Rightarrow 10 \frac{d^2x}{dt^2} = -2k \frac{dx}{dt} - 245x$ $\Rightarrow 10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 245x = 0 \text{ (Shown)}$
(ii)	$10 \frac{d^2x}{dt^2} + 80 \frac{dx}{dt} + 245x = 0$ Characteristic equation: $10m^2 + 80m + 245 = 0$ $\Rightarrow m = \frac{-80 \pm \sqrt{80^2 - 4(10)(245)}}{20}$ $\Rightarrow m = \frac{-80 \pm \sqrt{-3400}}{20}$ $\Rightarrow m = -4 \pm \frac{\sqrt{34}i}{2}$ $\therefore x = e^{-4t} \left(A \cos \frac{\sqrt{34}}{2} t + B \sin \frac{\sqrt{34}}{2} t \right).$ When $t = 0, x = 0.2, \frac{dx}{dt} = 3$. $0.2 = A \cos(0) + B \sin(0) \Rightarrow A = 0.2$

	$\frac{dx}{dt} = e^{-4t} \left(-\frac{\sqrt{34}}{2} A \sin \frac{\sqrt{34}}{2} t + \frac{\sqrt{34}}{2} B \cos \frac{\sqrt{34}}{2} t \right)$ $- 4e^{-4t} \left(A \cos \frac{\sqrt{34}}{2} t + B \sin \frac{\sqrt{34}}{2} t \right)$ $\Rightarrow 3 = \frac{\sqrt{34}}{2} B - 4(0.2) \Rightarrow B = \frac{19\sqrt{34}}{85}$ $\therefore x = e^{-4t} \left(\frac{1}{5} \cos \frac{\sqrt{34}}{2} t + \frac{19\sqrt{34}}{85} \sin \frac{\sqrt{34}}{2} t \right)$
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9	Suggested Solution
(i)	r represents the growth rate of the wrasse fish. M represents the carrying capacity of the fishery.
(ii)	$\frac{dW}{dt} = rW \left(1 - \frac{W}{16000} \right) - 200h$
(iii)	$\frac{dW}{dt} = 0.5W \left(1 - \frac{W}{16000} \right) - 200h$ $0.5W \left(1 - \frac{W}{16000} \right) - 200h = 0$ $\Rightarrow 0.5W - \frac{W^2}{32000} - 200h = 0$ $\Rightarrow W^2 - 16000W + 3200000h = 0$ Bifurcation occurs when $(16000)^2 - 4(3200000h) = 0$ $\Rightarrow h = 20$



(iv)

$$\frac{dW}{dt} = 0.5W \left(1 - \frac{W}{16000} \right) - 2000$$

$$0.5W \left(1 - \frac{W}{16000} \right) - 2000 = 0$$

$$\Rightarrow W^2 - 16000W + 32000000 = 0$$

To ensure sustained harvesting,

$$W \geq \frac{16000 - \sqrt{16000^2 - 4(32000000)}}{2}$$

$$\Rightarrow W \geq 2343.15$$

Thus the least initial number of wrasse fish required is 2344.